

★ PRIMER · WEEK 1

Before the Lecture: What a Model Is

Unit 1 · ML and the Finance Problem

Watch this first if machine learning is new to you

Twenty minutes of groundwork — the vocabulary, the three paradigms, and why we model returns rather than prices. If you have seen all of it, treat it as revision.

How to Use This Primer

Two audiences, one deck — read the column that describes you

If ML is new to you

- Watch before the lecture.
- Every assumed term is here.
- Do slide 8's arithmetic.

If you have done ML before

- Skim it — this is revision.
- Slide 7 is what finance changes.
- Slide 16 says if you can skip.

Nothing here is examined on its own — it is the floor the lecture stands on

Where Unit 1 Sits

The course builds one pipeline; this unit lays the first stone

Unit 1

you are here

What a model is, and why
finance is the hard case.

Unit 2

next

Rebuild the data itself — bars
instead of the clock.

Unit 3

then

Fix the label — what exactly are
we predicting?

Each unit adds one stage. Miss the first and the rest float free.

What a Model Actually Is

A model is a rule that turns what you know into a guess about what you do not.

The three parts

- Input — the features.
- Rule — a function with parameters.
- Output — the guess.

What learning means

- You do not pick the parameters.
- They are fitted to past data.
- Learning is finding that fit.

No magic. A model is arithmetic whose constants were set by data.

The Five Words the Lecture Assumes

Say these out loud once and the lecture stops being foreign

Word	What it means	In our first problem
Observation	one row of data	one trading day
Feature	a column you use as input	yesterday's return
Label	the answer you want to predict	did the price rise?
Training set	rows the model may learn from	2015–2022
Test set	rows kept hidden until the end	2023–today

A model that sees the test set during training has not been tested at all.

Three Ways a Machine Can Learn

Unit 1 names all three; units 4–8 use the first, unit 9 the second

Paradigm	You give it	It gives back	Example here
Supervised	inputs and correct answers	a rule that predicts the answer	will this loan default?
Unsupervised	inputs only	structure it found by itself	which stocks move together
Reinforcement	a goal and a sandbox	a policy — what to do when	how to execute a large order

Most of finance is supervised — but the labels are the hard part, as unit 3 shows.

Why We Model Returns, Not Prices

Prices wander without bound. Returns hover around zero — and that is what makes them modellable.

A price series

- Depends on all of history.
- Mean and variance drift.
- Two eras are not comparable.

A return series

- Each value is a % change.
- Roughly centred, comparable.
- The unit everything else uses.

This one substitution is the oldest idea in quantitative finance.

Work It Once by Hand

Five closing prices — compute the four returns yourself before reading on

Day	Close	Return = (today – yesterday) / yesterday	Value
Mon	100.0	—	—
Tue	102.0	$(102 - 100) / 100$	+2.00%
Wed	101.0	$(101 - 102) / 102$	-0.98%
Thu	104.0	$(104 - 101) / 101$	+2.97%
Fri	103.0	$(103 - 104) / 104$	-0.96%

Four returns from five prices — you always lose one row. Remember that.

Mean and Variance, From Those Four Numbers

Every risk measure in the course is built on these two

Mean — the centre

- Add the four: 3.03.
- Divide by 4: 0.76% a day.
- It says where returns sit.

Variance — the spread

- Centre each, square, average.
- Its root is the standard deviation.
- In finance: volatility.

Mean is the reward. Volatility is the risk. Everything else is detail.

What a Fat Tail Means

The problem set opens with this — here is the picture behind it

If returns were normal

- A 5-sigma day: once in 7,000 years.
- Risk models could be simple.
- Textbook statistics would work.

What markets actually do

- 5-sigma days arrive every few years.
- The extremes are where money is lost.
- That is what 'fat tail' names.

Fat tails are not an anomaly to be cleaned away — they are the subject.

Overfitting, in One Row of Pictures

The single most expensive mistake in this field

Too simple

underfit

A straight line through a curve.
Wrong on training and test
alike.

high bias

About right

the goal

Captures the shape, ignores the
noise. Works on data it never
saw.

Too complex

overfit

Threads every point, including
the noise. Perfect past, useless
future.

high variance

In finance the third one is seductive, because a perfect backtest looks like skill.

Bias and Variance — The Trade You Cannot Escape

The lecture states this formally; here is the plain version

	Bias	Variance
What it is	error from being too simple	error from chasing noise
Symptom	wrong even on training data	great on training, bad on test
Fix	a richer model	more data, or a simpler model
Finance's danger	rare — models are rarely too simple	everywhere — this is the default failure

Reduce one and the other grows. There is no setting where both vanish.

Notation You Will Meet on Wednesday

Four symbols carry most of the lecture

Symbol	Read it as	It stands for
x	ex	the input features of one observation
y	why	the true label you are trying to predict
$\hat{y}$	y-hat	the model's prediction — note the hat
θ	theta	the parameters the model learns

The hat always means 'estimated'. It is the difference between truth and guess.

The Python You Will See

You do not need to write it yet — you need to recognise it

Line	What it does
<code>import pandas as pd</code>	the table library — rows and columns
<code>df = pd.read_csv(path)</code>	load a CSV into a table called a DataFrame
<code>df['close'].pct_change()</code>	turn a price column into returns
<code>df.describe()</code>	mean, standard deviation, min and max at a glance
<code>train_test_split(X, y)</code>	cut the data into a training and a test set

Install nothing yet. Unit 1's lab walks the setup line by line.

Three Things Beginners Get Wrong

Each one costs a week if you carry it into unit 3

1

"More data always helps"

More of the same regime helps. Data from a market that no longer exists can hurt.

2

"A high accuracy means a good model"

Predict 'no default' every time and you are 97% accurate — and completely useless.

3

"The backtest proves it works"

A backtest proves it fitted the past. Unit 10 is entirely about telling the two apart.

If one of these surprised you, that is exactly what the primer was for

Self-Check Before Wednesday

Answer these three out loud — if you can, you are ready for the lecture

1	From 20 daily prices, how many returns do you get?	19
2	Which paradigm has no labels at all?	unsupervised
3	A model scores 99% on training and 51% on test. What is wrong?	overfit

Cover the right column, answer, then check. All three come from this deck.

What Wednesday's Lecture Does With This

You bring the vocabulary; the lecture brings the argument

1

It makes the finance case specific

Why the signal is faint, the data non-stationary, and the labels overlapping.

2

It contrasts two ways of working

The Sisyphus strategy against the meta-strategy — process against genius.

3

It ends in code

The problem set works fat tails, overfitting and credit default by hand, then in Python.

Next: Unit 1 lecture — The World of ML and the Finance Problem